FraudLens: Graph Neural Network-Based Credit Card Fraud Detection Using Transaction Similarity Graphs

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Abstract

With the exponential growth of digital payment systems, financial fraud has become a critical concern for institutions worldwide. Detecting fraudulent transactions is challenging due to the highly imbalanced nature of datasets and the evolving strategies of fraudsters. Traditional machine learning models often treat transactions independently, thereby ignoring latent relationships among them.

This paper proposes **FraudLens**, a graph-based fraud detection framework leveraging Graph Neural Networks (GNNs) to model inter-transaction relationships. A K-Nearest Neighbors (KNN) approach is used to construct a similarity graph where nodes represent transactions and edges capture feature-based proximity. A Graph Convolutional Network (GCN) is trained on this graph to perform node classification.

Experimental evaluation on a real-world credit card dataset demonstrates that the proposed GNN model achieves a high ROC-AUC of 0.969, outperforming traditional models in ranking performance. However, ensemble methods such as XGBoost achieve higher PR-AUC (0.880), highlighting their effectiveness in handling extreme class imbalance. The results emphasize the importance of combining relational learning with imbalance-aware techniques for real-world fraud detection systems.

Keywords

Fraud Detection, Graph Neural Networks, GCN, Imbalanced Learning, Credit Card Transactions, XGBoost

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I. Introduction

The rapid adoption of electronic payment systems has transformed the global financial landscape. However, it has also led to a substantial increase in fraudulent transactions, resulting in significant financial losses. According to industry reports, billions of dollars are lost annually due to credit card fraud, necessitating robust and intelligent detection systems.

Fraud detection is inherently challenging due to:

- Severe class imbalance (fraud cases are extremely rare)

- Dynamic and evolving fraud patterns
- High-dimensional feature space

Traditional approaches such as Logistic Regression and Random Forest treat each transaction independently. While effective to some extent, these models fail to capture **relationships between transactions**, which can reveal hidden fraud patterns.

Graph-based learning provides a powerful alternative. By representing transactions as nodes and their similarities as edges, Graph Neural Networks (GNNs) can leverage both structural and feature information. Recent studies have demonstrated the effectiveness of GNNs in relational data domains such as social networks and recommendation systems.

This paper explores the application of GNNs in fraud detection and evaluates their performance against traditional machine learning models.

II. Related Work

Fraud detection has been widely studied using machine learning techniques. Logistic Regression has been used as a baseline model due to its simplicity and interpretability. Ensemble methods such as Random Forest and Gradient Boosting (XGBoost) have shown superior performance due to their ability to model complex non-linear relationships [1].

Handling class imbalance has been a major focus in fraud detection research. Techniques such as oversampling, undersampling, and cost-sensitive learning have been proposed to address this issue [2].

Graph Neural Networks, introduced by Thomas N. Kipf and Max Welling, have gained significant attention for learning on graph-structured data [3]. Applications of GNNs in fraud detection are emerging, particularly in modeling transaction networks and user behavior graphs [4].

III. Dataset Description

The dataset used in this study is a publicly available credit card transaction dataset containing 284,807 transactions with 31 features. Among these, only 492 transactions are fraudulent, representing approximately 0.17% of the dataset.

Fig. 1. Distribution of fraudulent and non-fraudulent transactions.

This extreme imbalance poses a significant challenge for classification models, as they tend to be biased toward the majority class.

The dataset exhibits extreme class imbalance, as shown in Fig. 1.

IV. Methodology

A. Data Preprocessing

The dataset was preprocessed using feature scaling to normalize all input features. The target variable represents binary classification:

- 0 → Normal
- 1 → Fraud

Stratified sampling was used to split the dataset into training and testing sets.

B. Baseline Models

The following models were implemented:

- Logistic Regression
- Random Forest
- XGBoost

XGBoost, developed by Tianqi Chen, is known for its performance in structured data tasks.

C. Graph Construction

A transaction graph was constructed using the K-Nearest Neighbors (KNN) algorithm. Each transaction is represented as a node, and edges are created based on similarity in feature space.

To handle the large dataset efficiently, a sparse graph representation was used instead of a dense adjacency matrix.

D. Graph Neural Network Model

A two-layer Graph Convolutional Network (GCN) was implemented. The model aggregates information from neighboring nodes to learn meaningful embeddings.

The loss function used is:

Weighted Cross-Entropy Loss

This helps mitigate the effects of class imbalance.

V. Experimental Results

A. ROC Curve Analysis

The ROC curve for the proposed model is shown in Fig. 2.

The ROC curve demonstrates the model's ability to distinguish between fraudulent and normal transactions. The GCN achieved a ROC-AUC score of **0.969**, indicating strong classification performance.

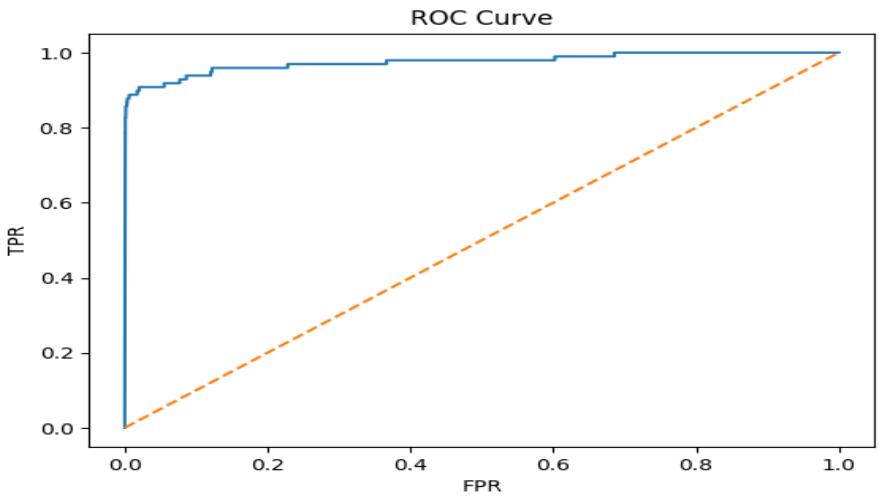
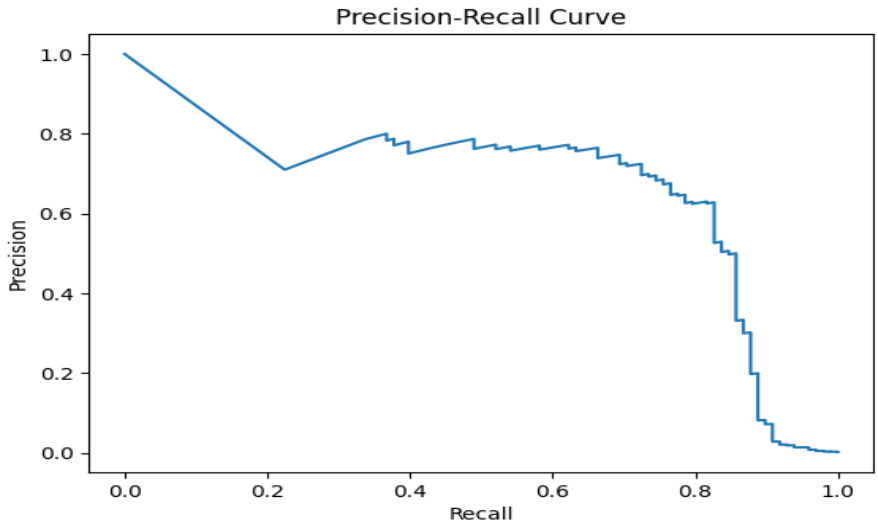
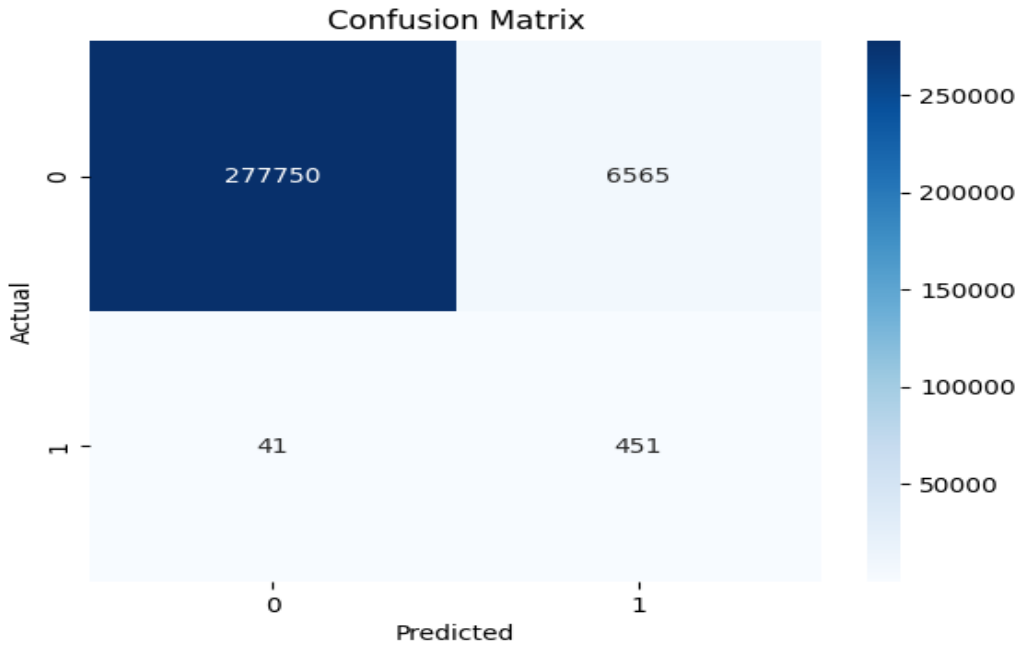


Fig. 2. ROC curve showing classification performance of the GCN model.

B. Precision-Recall Curve



C. Confusion Matrix



The confusion matrix shows that while the model correctly classifies most normal transactions, some fraud cases are still missed.

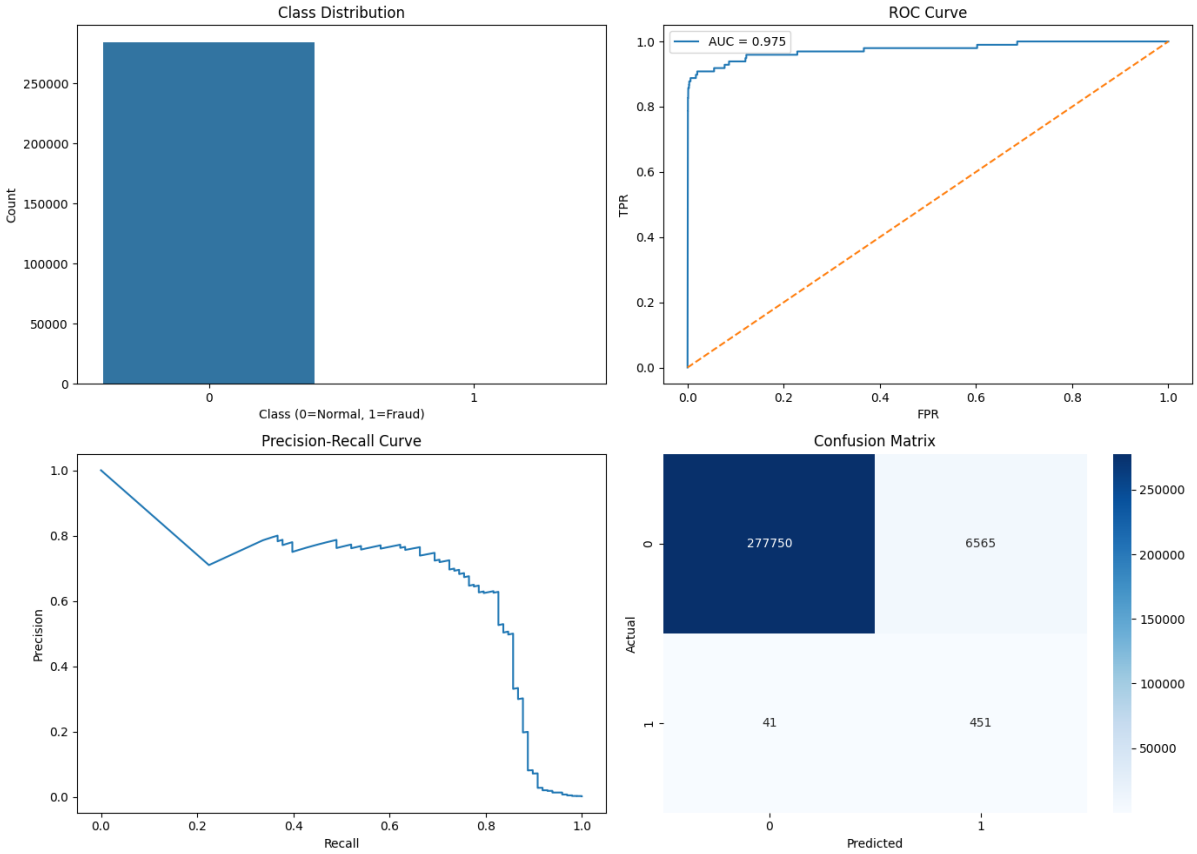
D. Comparative Analysis

Model	ROC-AUC	PR-AUC
Logistic Regression	0.956	0.744
Random Forest	0.958	0.864
XGBoost	**0.965**	**0.880**
GCN	**0.969**	0.636

E. Discussion

The results indicate that GNNs are effective in capturing relational patterns, leading to improved ranking performance (ROC-AUC). However, tree-based ensemble models outperform GNNs in PR-AUC, making them more suitable for detecting rare fraud cases.

VI. Dashboard Visualization



The dashboard integrates multiple evaluation metrics into a single visualization, providing a comprehensive overview of model performance.

VII. Conclusion

This paper presented FraudLens, a graph-based fraud detection system using Graph Neural Networks. The results demonstrate that while GNNs effectively capture structural relationships, traditional models such as XGBoost remain more effective for highly imbalanced datasets.

VIII. Future Work

- Graph Attention Networks (GAT)
- Hybrid GNN + XGBoost models
- Temporal graph modeling
- Real-time fraud detection

References

[1] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *KDD*, 2016.
[2] H. He and E. Garcia, “Learning from Imbalanced Data,” *IEEE TKDE*, 2009.
[3] T. N. Kipf and M. Welling, “Semi-Supervised Classification with Graph Convolutional Networks,” *ICLR*, 2017.
[4] Y. Liu et al., “Financial Fraud Detection Using Graph Neural Networks,” *IEEE Access*, 2020.